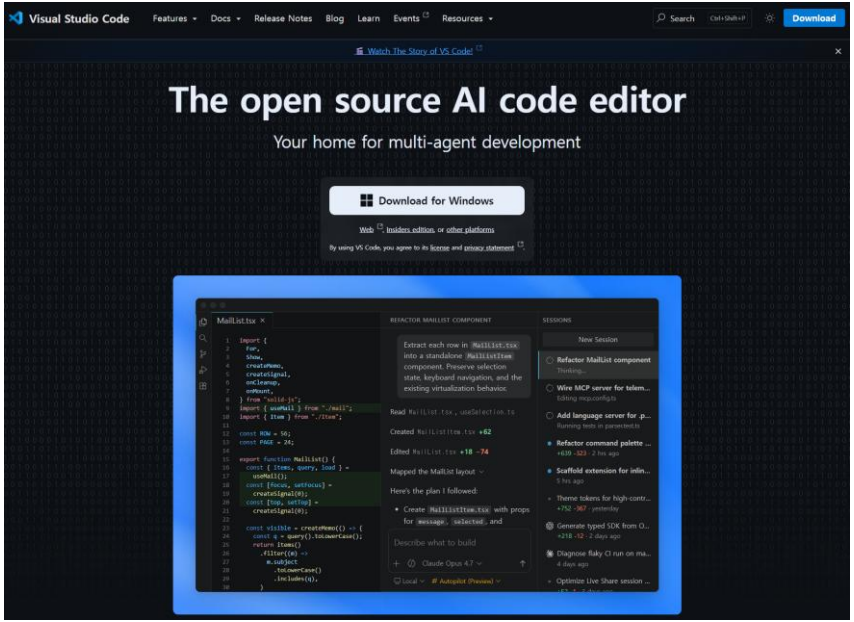
Prerequisite1 : Download VSCode

- **Click the Link Below**
  - https://code.visualstudio.com/



Prerequisite2 : Download Python

- **Click the Link Below**
  - https://www.python.org/downloads/
  - Around 3.11 is recommended

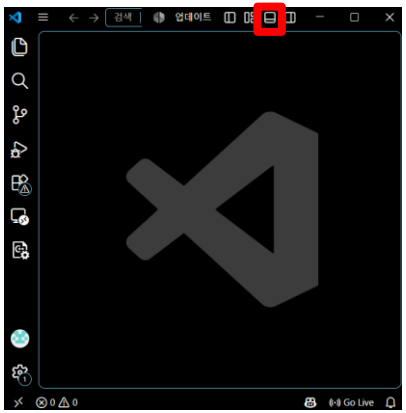


Prerequisite3 : Environmental Settings

☐ Write the Commands into Your Terminal

- *pip install pandas ← Enables csv manipulation*
- *pip install sqlglot ← Enables SQL syntax validation*

Open terminal

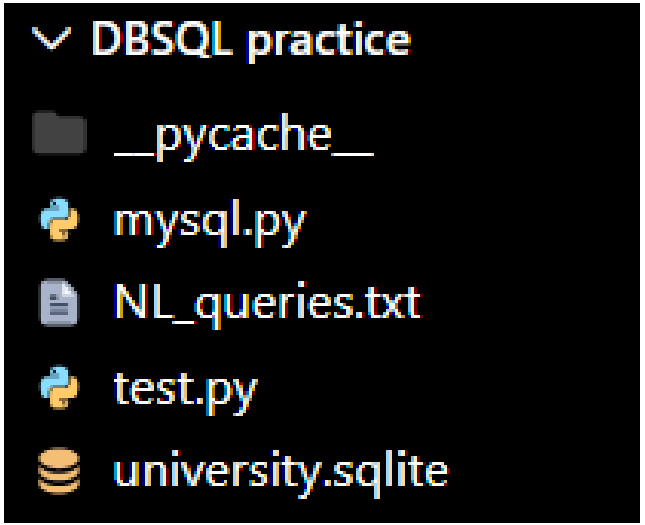


Example terminal output when sqlglot is activated

```
[SQL Syntax Error]
Invalid expression / Unexpected token. Line 14, Col: 23.
  Id
GROUP BY
    d.driverId,
    d.forename,
    d.surname
HAVING COUNT(*) >= 20
ORDER BY race_count DEC;
```

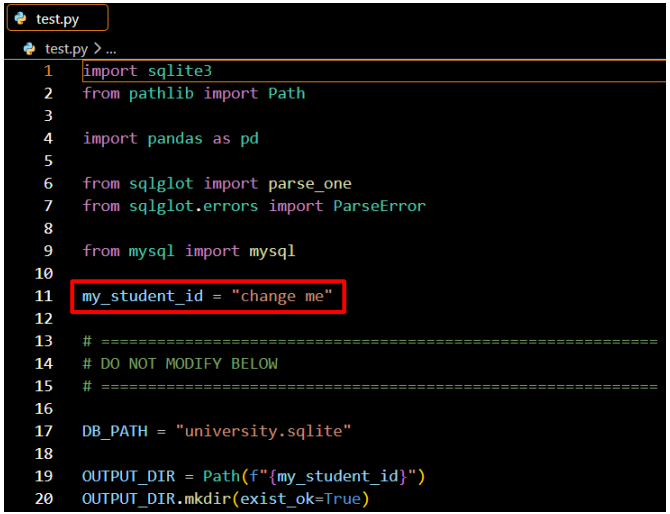
Step1 : Download Skeleton Code

- **Download Folder from Eclass**
  - Un-zip it and open it in your editor



Step2 : test.py

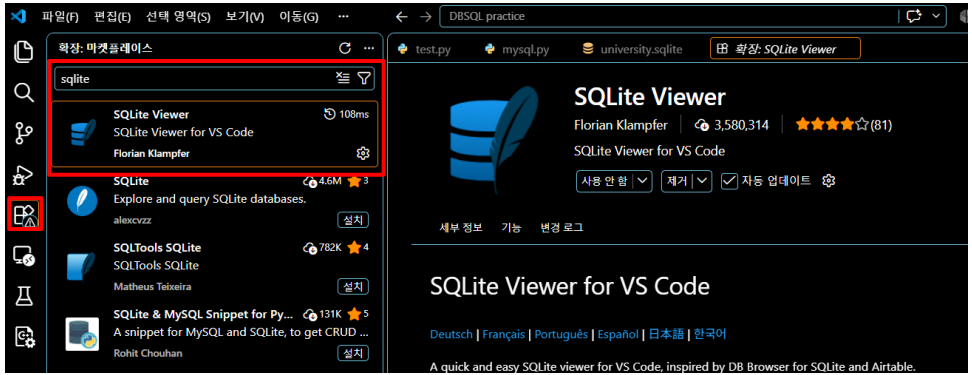
- Replace “change me” to Your “Student ID”
  - Do not modify test.py elsewhere
  - Further modification may lead to inconsistent output and affect grading

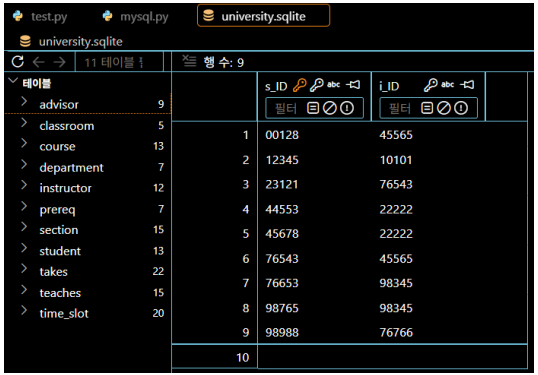


Step3 : university.sqlite

☐ Directly See How the DB is Structured with SQLite Viewer

- This is optional for your conveniency





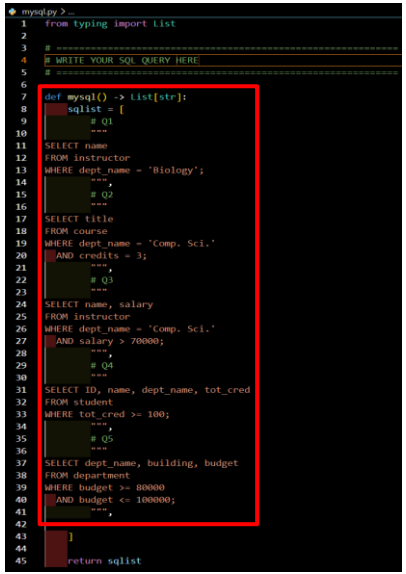
Step4 : NL_queries.txt & mysql.py

- **Write Your SQL to mysql.py based on NL_queries.txt**
  - We already wrote golden answers corresponding to NL_queries.txt
  - Nothing more to do here
  - mysql.py will be empty for upcoming assignments

NL_queries.txt

```
1. Find the names of all the instructors from Biology department.
2. Find the names of courses in Computer science department which
3. Find the names and salaries of instructors in the Comp. Sci. de
4. Find the ID, name, department name, and total credits of all st
5. Find the department name, building, and budget of all departmen
```

NL_queries.txt



```
from typing import List

# ============================================
# WRITE YOUR SQL QUERY HERE
# ============================================

def mysql() -> List[str]:
    sqlist = [
        # Q1
        """
SELECT name
FROM instructor
WHERE dept_name = 'Biology';
        """,
        # Q2
        """
SELECT title
FROM course
WHERE dept_name = 'Comp. Sci.'
  AND credits = 3;
        """,
        # Q3
        """
SELECT name, salary
FROM instructor
WHERE dept_name = 'Comp. Sci.'
  AND salary > 70000;
        """,
        # Q4
        """
SELECT ID, name, dept_name, tot_cred
FROM student
WHERE tot_cred >= 100;
        """,
        # Q5
        """
SELECT dept_name, building, budget
FROM department
WHERE budget >= 80000
  AND budget <= 100000;
        """,

    ]

    return sqlist
```

mysql.py

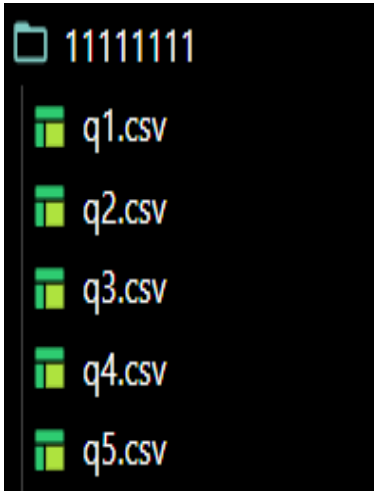
Step5 : test.py

Check the SQL Outputs

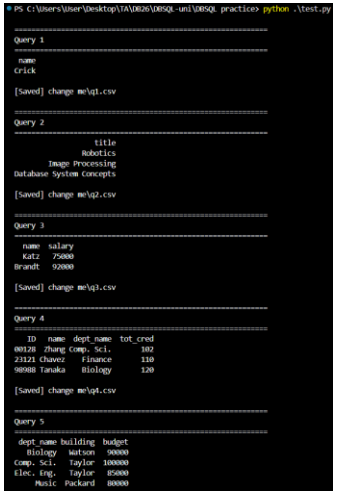
- Run test.py to see the results

```
PS C:\Users\User\Desktop\TA\DB26\DBSQL-uni\DBSQL practice> python .\test.py
```

Output 1



Output 2



```
PS C:\Users\User\Desktop\TA\DB26\DBSQL-uni\DBSQL practice> python .\test.py
================================================================
Query 1
================================================================
 name
Crick

[Saved] change me\q1.csv

================================================================
Query 2
================================================================
                title
             Robotics
      Image Processing
Database System Concepts

[Saved] change me\q2.csv

================================================================
Query 3
================================================================
  name  salary
  Katz   75000
Brandt   92000

[Saved] change me\q3.csv

================================================================
Query 4
================================================================
   ID   name  dept_name  tot_cred
00128  Zhang Comp. Sci.       102
23121 Chavez    Finance       110
98988 Tanaka    Biology       120

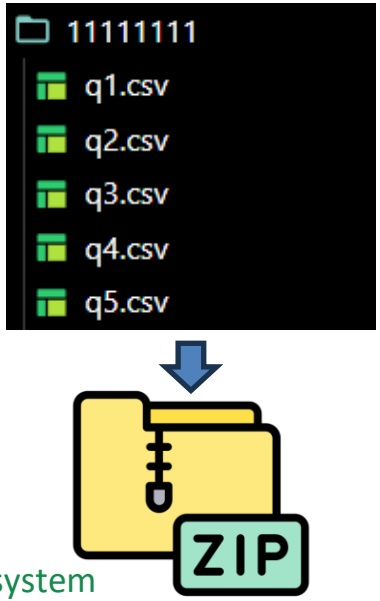
[Saved] change me\q4.csv

================================================================
Query 5
================================================================
 dept_name building  budget
   Biology   Watson   90000
Comp. Sci.   Taylor  100000
Elec. Eng.   Taylor   85000
     Music  Packard   80000
```

Submission Guideline

Submission

- What: **A Compressed results folder**
  - Compress your {student_id} folder with the queried answers in it
  - Compressed folder name should be just {student_id}
- Where: Assignment board on e-class
- Deadline: End of class (tentative)



Q&A

- Joohee Cho (조주희): joohee2008@cau.ac.kr
- Sooyong Jung (정수용): wjdtndyd2001@cau.ac.kr
- Sooho Moon (문수호): moonwalk725@cau.ac.kr

(Bonus) Usage of Skeleton Code

- Future assignments will operate under the same code format and submission system
- Students can practice SQL for future exams/assignments